Counterfactual Concept Swaps on InceptionV1 Curve Detectors: a Specific but Partial Causal Mediator of Curve Orientation

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Abstract

Circuits-style neuron explanations in vision models rest mostly on observational evidence. We test one of the best-characterised, "the ten InceptionV1 `mixed3b` units are curve detectors tuned to orientation", with pre-registered counterfactual interchange interventions (concept swaps). After reproducing the authors' published tuning curves ($r = 0.969$), we copy the family's activations between images that differ only in curve orientation and read the next layer's published curve units. Compared with other channels, the family is a highly *specific* mediator: it moves the next layer's curve code 0.410 of the way to the source, versus a 95th percentile of 0.098 for 2000 activation-change-matched 10-channel controls, and ranking all 480 channels by their single-channel patching effect (post hoc) puts 9 of the 10 published units in the top 10 for the next layer's curve units (8 for a decoder over the whole next layer). It carries curvature evidence: inserting its arc activations into a line image makes the next layer's curve units read "arc" in 0.807 of pairs (post-hoc matched-control 95th percentile 0.114). Its ablation changes the network's predictions on natural images about 1.8 times as often as matched ablations (0.172 vs 0.093; exploratory). It is *not sufficient*: the next layer decodes the source orientation in only 0.177 of pairs [0.156, 0.199], so the pre-registered strong hypothesis is falsified. The rest of the orientation signal is spread over many channels (about 80 for reliable transfer under fixed per-channel rankings), and whether they can stand in for the family depends on the readout: the other 470 channels transfer the orientation in 0.403 of pairs for the next layer's curve units, but in only 0.019 for a decoder over the whole next layer, where the family carries 0.573 of the decoded shift. The same specific-but-partial pattern holds for the 3a→3b curve circuit, on natural backgrounds, and under a natural rotation counterfactual. Post-hoc checks leave these conclusions unchanged: a stricter rank-matched null, a causally ranked competitor set, per-direction curvature nulls, a cross-nuisance sham, the decoded-orientation metric, and blur-free rotations.

1 Introduction

Neuron-level explanations of vision models are mostly *observational*. A unit is called a "curve detector" because it fires on curves, because its feature visualization looks like a curve, and because its weights connect to other curve-like units [7, 8, 26]. The interventional evidence is circuit (weight) editing and a preliminary experiment that removes the 3b curve detectors and measures top-5 accuracy [8]. In language models, Huang et al. [19] showed that even the most confident GPT-4-generated explanations of GPT-2 XL neurons have high error rates and little to no causal efficacy when tested *interventionally*. In their test, the explanation is a causal claim that the unit mediates the concept: copying the unit's value from a source input into a base input should make the model behave as if it saw the source. Interventional tests now routinely validate representations in language

models: interchange interventions on digit-position neurons [3], ablation and steering of SAE and attribution-graph features [2, 6], and concept swaps of Jacobian-lens vectors [17].

We apply this interventional standard to one of the best-characterised neuron families in InceptionV1: its ten `mixed3b` curve detectors [7, 15]. The explanation names a concrete variable, curve orientation. That makes exact counterfactual pairs possible: two images identical except for orientation. It also lets us check our setup against the authors' published tuning data before any intervention.

Contributions.

- We reproduce the published tuning curves of the 3a, 3b and 4a curve families. The fit is $r = 0.969$ for 3b and $r = 0.847$ for 4a, with the orientation convention fitted on 3a only (§4.1).
- We run pre-registered interchange interventions with activation-change-matched and branch-matched null neuron sets. Readouts are the published `mixed4a` curve family and, separately, all of `mixed4a`. We also test natural images.
- The main result is mixed. The curve family moves the next layer's orientation code 0.410 of the way to the source; 95% of activation-change-matched 10-channel controls move it less than 0.098, and the best of them transfers the decoded orientation in 0.13% of pairs. Yet swapping the family makes the next layer decode the source orientation in only 0.177 of pairs. By our pre-registered rule, the strong sufficiency claim is falsified. The family is a highly specific but partial mediator. The other `mixed3b` channels carry the rest of the orientation signal, and how far they can replace the family depends on the readout.

2 Background and related work

Circuits and curve detectors. Olah et al. [26] introduced the circuits framing, and Olah et al. [27] catalogued InceptionV1's early-vision families. Cammarata et al. [7] characterised ten `mixed3b` curve units with synthetic curves and radial tuning curves. Cammarata et al. [8] reverse-engineered how 3a curves build 3b curves and validated parts of the circuit by editing weights. In a preliminary experiment, removing the 3b curve detectors lowered top-5 accuracy on 2000 ImageNet validation images from 88.6% to 86.3%, and artificial curve detectors in their place recovered 87.6%. Olah et al. [28] described the family as rotated copies of one feature, and Voss et al. [32] found all curve-related `mixed3b` features in its 5×5 branch. Gorton [15] found additional curve features with sparse autoencoders. Cammarata [9] revisits layer 3b and argues that neurons are cleaner than superposition accounts suggest. None of these works runs counterfactual activation swaps on the family.

Evaluating neuron explanations. Network dissection labels units [4], and later work ablates and inserts them in classifiers and GANs [5]. Class selectivity is a poor predictor of ablation importance [23], regularising it away does not hurt accuracy [21], and feature visualizations can mislead [14]. Synthetic-image checks [20] and metric sanity checks [24] evaluate explanations observationally. Recent work frames unit interpretation as falsifiable hypothesis testing [1].

Interchange interventions. Causal abstraction [11, 13] formalises when a set of units realises a high-level variable. Its operational test is the interchange intervention, also called activation patching. Metric choice can change patching conclusions [33], so next to Huang et al.'s binary

success measure (IIA) we report a continuous, roughly linear effect, as Heimersheim and Nanda [18] advise. Resampling-based hypothesis tests [10] motivate our complement and null sets.

3 Method

Model. We use the Lucid InceptionV1 ("inception5h") [31], loaded with Lucent. Its local response normalisation is set to the TensorFlow-exact constants (window $11 = 2 \cdot \text{depth_radius} + 1$, $\alpha = 11 \cdot 10^{-4}$, $\beta = 0.5$, $k = 2$), which follow from the graph's LRN attributes and TensorFlow's LRN formula.[1] A third-party comment on Lucent PR #51 gives the same derivation and reports a largest normalised error of $6 \cdot 10^{-6}$ against the original TensorFlow graph; the PR itself proposes a window of 10. All convolutions run in fp32 with TF32 disabled.

Stimuli and splits. Arcs are rendered with 4× supersampling. Each spans 120°, and its midpoint sits on the receptive-field (RF) centre of the readout units, pixel (111,111), found from the input-gradient centre of mass [22]. The orientation θ is the direction from the midpoint to the centre of curvature. A split is 72 orientations × radii $\{30, 45, 60, 80\}$ × widths $\{3, 5\}$ × two polarities, giving 1152 images. Each nuisance cell gets one random translation of ± 2 px, and the jitter seed defines the split: selection (seed 1) and confirmatory (seed 2). Each base gets four sources from its own cell with $|\Delta\theta| \geq 30°$, giving 4608 minimal pairs.

Intervention. The readout is `mixed4a` unit (6,6). It depends only on an 11×11 window of `mixed3b`: the 5×5 convolution sees `maxpool4` cells 4–8, which pool `mixed3b` rows 8–18. For a neuron set S we overwrite channels S of the base window with the source's values and recompute `maxpool4`→`mixed4a` on the window. In fp64 this equals the same block run on the full map (the unit test asserts agreement to 10^{-9}; a re-check measured a difference of 0), and in fp32 that block matches Lucent's forward pass within 10^{-4}.

Readouts τ and metrics. The primary readout uses the 14 `mixed4a` curve units in the authors' published tuning data (the article's figure labels 12), read pre-ReLU and z-scored on the selection split. Orientation is decoded by maximum Pearson correlation with the 72 clean selection templates of the pair's nuisance cell. The interchange intervention accuracy is

$$\mathrm{IIA}(S) = \frac{1}{|P|} \sum_{(b,s)\in P} \mathbf{1}\Big[\big|\tau(M_{S\leftarrow s}(b)) - \tau(M(s))\big|_\circ \leq 10°\Big], \tag{1}$$

taken over eligible pairs P: those whose clean base and clean source both decode within 10° of their true orientation. The continuous *normalized effect* is $\sum\langle z_p - z_b, z_s - z_b\rangle / \sum\|z_s - z_b\|^2$, over pairs outside the lowest 5% of $\|z_s - z_b\|^2$. It is 0 for no change and 1 when the readout becomes the source's. The secondary, non-circular readout is a ridge regression from all 508 `mixed4a` channels to $(\cos\theta, \sin\theta)$.

Controls. A patch acts only through channels whose values differ between base and source, so a random null is too easy to beat. Our primary null draws 2000 non-family sets matched to the family's histogram over deciles of Δ_c, the mean squared base→source change of channel c on selection pairs. A co-primary null draws 2000 sets from the rest of the 5×5 branch. References are the Fisher-score top-10 channels for θ [3, 16], the complement (470 channels) and all 480 channels.

[1] `https://www.tensorflow.org/api_docs/python/tf/nn/local_response_normalization`; comment: `https://github.com/greentfrapp/lucent/pull/51#issuecomment-6345247394`

Statistics. Pairs share base and source images, so we use the pigeonhole bootstrap [29] ($B = 2000$), recomputing the null's 95th percentile in every replicate. p-values are $(1+\#\{\text{null} \geq \text{obs}\})/(1+M)$ [30], and secondary tests are Holm-corrected.

Pre-registered decision rule. Written before the confirmatory run, with pilot results disclosed. Let $D = \text{IIA}_{\text{family}} - q_{95}(\text{null})$. H1 is *supported* if IIA ≥ 0.5, $D \geq 0.10$, and the 95% CI lower bound of D is above 0 against both nulls. It is *falsified* if the CI upper bound of D is below 0.05 or IIA < 0.2; otherwise it is inconclusive. Each secondary hypothesis is supported if its normalized-effect margin over the Δ-matched q_{95} has a CI lower bound above 0 and a Holm-adjusted $p < 0.05$.

4 Results

4.1 E1: the observational explanation replicates

We fitted one orientation convention (a reflection plus a 180° shift) on `mixed3a` alone, with mean $r = 0.937$. Under that convention, our radial tuning curves match the published ones with mean Pearson $r = 0.969$ for the 10 `mixed3b` units (every unit between 0.956 and 0.980) and $r = 0.847$ for the 14 `mixed4a` units. The 95th percentiles of a unit-label permutation null are 0.254 and 0.242, and $p = 10^{-4}$ for each (Fig. 1). Each 3b unit separates arcs within ±45° of its preferred orientation from other orientations with AUROC 0.70–0.95, and from straight lines with AUROC 0.67–0.92. AUROC is descriptive only: unlike Pearson correlation, it fails one of the sanity checks of Oikarinen et al. [24]. With Lucent's default LRN constants instead of the TF-exact ones, the correlations are essentially unchanged (0.969 for 3b, 0.848 for 4a). Our model port, renderer and readout therefore reproduce the evidence the explanation rests on. This is a descriptive check without a pre-registered threshold. Because the stimulus radii were chosen on the pilot split to maximise this agreement, the r values are somewhat optimistic; on the pilot, 3b r was at least 0.929 at every radius from 20 to 80 px.

4.2 E2: the family is specific but not sufficient

H1 (primary). Clean decoding accuracy is 0.996, leaving 4574 of 4608 pairs eligible. Swapping the 10 curve units yields IIA = 0.177 [0.156, 0.199]. This is below the pre-registered falsification bound of 0.2, so **H1 is falsified**. The verdict is identical for two further jitter seeds (IIA = 0.172, 0.171) and with Lucent's default LRN (0.172).

Specificity. The same swap is far more potent than any matched control (Fig. 2). The Δ-matched null has IIA $q_{95} = 0.001$ and the branch-matched null $q_{95} = 0.000$; across 4000 sets the best transfers orientation in 0.13% of pairs ($p = 0.0005$ against each null). The normalized effect of the family is 0.410 [0.400, 0.420], against a Δ-matched q_{95} of 0.098 and a branch-matched q_{95} of 0.077. The observational Fisher top-10, which uses only the window centre and shares 4 units with the family, reaches only IIA = 0.028; a causal ranking is the stronger competitor (§4.4). Mean-ablating the family leaves the base orientation decodable in 0.806 of pairs, against 0.985 for Δ-matched ablations and (post hoc) 0.982 for rank-matched ones. "Specific" here means specific relative to other `mixed3b` channels. A same-orientation sham leaves it decodable in 0.999.

Shared code and additivity. The 470 complementary channels reach IIA = 0.403 and effect 0.590. The two effects sum to 1.000. Over random K-subsets of the family, the effect grows almost linearly: $0.040, 0.080, 0.119, 0.202, 0.285, 0.410$ for $K = 1, 2, 3, 5, 7, 10$. IIA, in contrast, stays near 0

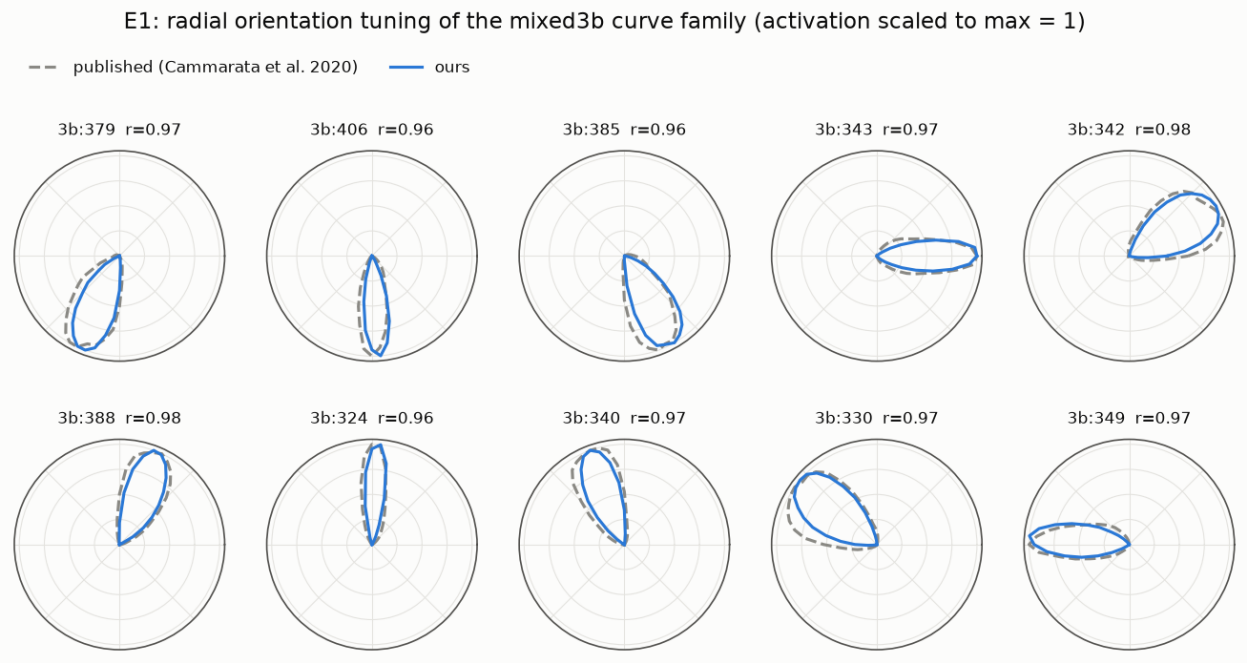


Figure 1: E1. Radial orientation tuning of the 10 `mixed3b` curve units: ours (blue) and published by Cammarata et al. [7] (gray, dashed), each scaled to a maximum of 1. The orientation convention was fitted on `mixed3a` only.

until $K = 7$ (0.045) and reaches 0.177 at $K = 10$. This additivity is only partly structural. `maxpool4` acts per channel, and in the 1×1 and pool branches of `mixed4a` a linear 1×1 convolution follows, so patches add exactly there; 6 of the 14 readout units are 1×1-branch units. The 3×3 and 5×5 branches mix channels in a bottleneck before a ReLU. On our stimuli their outputs are non-additive by 0.095 and 0.070 (squared-norm ratio), and the 8 readout units in the 5×5 branch by only 0.018 (post hoc). For the curve readout, near-additivity is therefore partly an empirical property of these stimuli, not of the architecture. Patching the top-K channels by activation change gives IIA $= 0.359, 0.442, 0.792, 0.979$ for $K = 20, 40, 80, 160$, and the 10 curve units are all among the top 17. Ranking channels by single-channel patching effect instead (post hoc) gives $0.485, 0.693, 0.809$ for $K = 20, 40, 80$. Reliable transfer (IIA ≈ 0.8) thus takes about 80 channels under either ordering. Both are fixed rankings by a per-channel score rather than a search over sets, so this bounds the smallest sufficient set from above. Whether the other channels can replace the family depends on the readout. Under the whole-layer ridge readout, the complement reaches only IIA $= 0.019$ (family 0.105), and in that decoder's $(\cos\theta, \sin\theta)$ output it carries 0.427 of the displacement against 0.573 for the family (post hoc).

Secondary tests. All five secondary hypotheses meet the pre-registered criterion. The Holm-adjusted p-values are ≤ 0.004 (Table 1; Fig. 3).

- **Ridge readout over all 508 `mixed4a` channels.** This is non-circular. As in the registered code (the pre-registration does not name the space), the effect is the projection over all 508 z-scored channels: 0.248 against a null q_{95} of 0.106, with IIA $= 0.105$. It therefore measures whole-layer movement. Post hoc, we also measure it on the decoded orientation (§4.4).
- **Curvature swap (H2).** The family's effect is 0.393 against 0.141. Inserting the family's arc

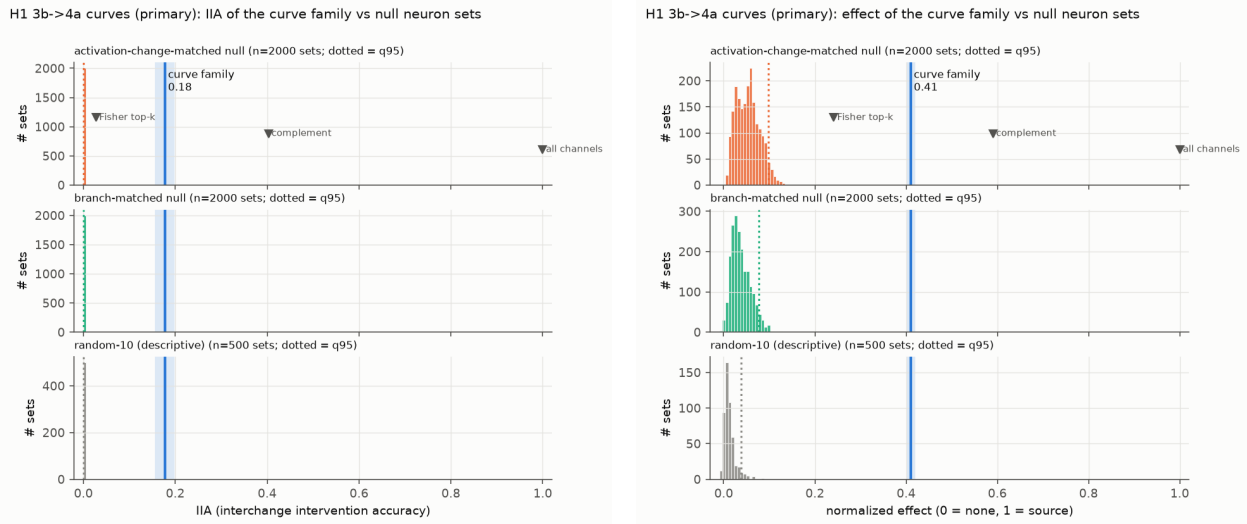


Figure 2: H1. IIA (left) and normalized effect (right) of the curve family (blue line, 95% CI band) against 2000 Δ-matched, 2000 branch-matched and 500 random 10-channel null sets. Triangles mark the reference sets.

activations into a line image makes the 4a curve units read "arc" in 0.807 of pairs. Against that direction's own Δ-matched null (post hoc), the null mean is 0.030 and its q_{95} is 0.114 ($p = 0.001$). The decision is binary (line vs arc template), a lower bar than H1's 72-way decode. The reverse direction succeeds in only 0.003 of pairs, yet its continuous effect, 0.367 against a null q_{95} of 0.130, is close to the line→arc effect of 0.418. The asymmetry in IIA therefore most likely reflects where the correlation decoder puts its line/arc boundary, not the family. This is an inference from the average effects; no analysis measures the boundary. In a simplified one-dimensional picture along the line-to-arc direction, the boundary would sit less than halfway from the line template, but per-pair moves vary and the decoder is not a threshold on that one direction.

- **The Curve Circuits claim, 3a→3b (H4).** The effect is 0.324 against 0.169. IIA is only 0.058 while the complement reaches 0.757, the same specific-but-partial pattern as H1.

In the exploratory analyses, IIA depends on the orientation difference: 0.111 for $|\Delta\theta|$ in 30–60° and 0.303 in 120–150°. A *family rotation* swap, a channel reassignment within the family inspired by the concept swaps of Gurnee et al. [17], gives each unit the map of the unit k steps further around the orientation circle. The circular-mean shift of the decoded orientation points in the predicted direction, but for $k \neq 5$ it reaches only 21–39% of the predicted angle (e.g. −8° vs −36° for $k = 1$). This mean mixes outcomes: 34–74% of pairs keep their decoded orientation, and 10–49% land within 30° of the predicted shift.

4.3 E3: natural images

With arcs composited onto Imagenette photos, the H1 pattern recurs: IIA = 0.190 and effect 0.415 against a null q_{95} of 0.104. In the natural rotation counterfactual (H5) on 506 curve-present photos, the patched base is read as the rotated version in IIA = 0.072 of pairs. Its effect is 0.287 against a null q_{95} of 0.075, while the complement dominates at IIA = 0.847. On curve-absent crops the family's effect is 0.135 (exploratory, without a null; the family is often silent there, so a smaller

effect is expected). End to end (a pre-registered exploratory analysis), mean-ablating the family over the whole map changes the network's 1008-way top-1 prediction on a fraction 0.172 of 1962 photos. Variance-matched null ablations change it on 0.093 (q_{95} 0.110, $p = 0.0099$), and the mean KL is 0.2054 against 0.0501. The null is matched in ablation size: the family's mean activation variance is 0.96 times that of the null sets (post hoc). This agrees with the preliminary removal experiment of Cammarata et al. [8], which had no matched null.

4.4 Post-hoc robustness (not pre-registered)

After the confirmatory run, an independent code review raised three ways the pre-registered comparisons could flatter the family. We tested each.

- **A softer-than-intended null.** All 10 curve units lie above the 96th percentile of activation change, so decile matching mostly draws channels that change less. A rank-matched null pairs each unit with one of its 10 nearest non-family channels by activation change (1000 sets). Against it, H1's effect 0.410 still far exceeds the null q_{95} of 0.121. The 10 non-family channels with the largest activation change reach only 0.125, and transfer the decoded orientation in 1 of 4574 pairs. All other tests keep margin CIs above 0 (`reports/posthoc_robustness.md`).

- **Z-space vs decoded orientation.** Measured in the ridge decoder's $(\cos\theta, \sin\theta)$ output space, the family moves the decoded orientation vector 0.573 of the way to the source's (rank-null q_{95} 0.126). It lands within 10° in only 0.105 of pairs, so the decoded orientation rotates only part of the way.

- **Rotation blur in H5.** Rotating square crops by $-90/90/180°$ is an exact transpose with no interpolation. Under these rotations the H5 effect is 0.336 (null q_{95} 0.077) and IIA 0.108. Both are larger than the pre-registered H5 values, but the third audit (below) showed that the difference comes from the angle set, not from blur: the exact run has no $\pm 45°$ rotations. On the same angles, the original rotations give effect 0.335 and IIA 0.108 (difference CI [-0.001, 0.001]).

A second audit of the finished study, run as a separate session, found no methodological error but four claims that lacked their own test: the family beats a data-driven set, H2's line→arc rate exceeds its own null (see *Curvature swap*), ablating the family disrupts orientation more than matched channels, and the rest of the code is redundant. The outcome that would overturn each claim was set in the audit plan before the run, but was first committed together with the results, so git does not timestamp it (`reports/posthoc_robustness.md`). The audit also added a cross-nuisance sham and a check of the relevance null's ablation size, which have no overturning rule.

- **A causal competitor set.** We rank every `mixed3b` channel by its single-channel patching effect on the selection split. The top 10 contain 9 family units; all family units rank within the top 12. The causal top-10 reaches effect 0.420 on the confirmatory pairs (family 0.410). The 10 best non-family channels reach only 0.201 (IIA = 0.001); the family's margin over them has CI [0.197, 0.221].

- **Redundancy under the whole-layer readout.** See *Shared code and additivity*: with the ridge decoder, the complement carries 0.427 of the decoded-orientation shift and transfers it in 0.019 of pairs.

- **Concept specificity (no overturning rule).** Copying the family from a stimulus with the same orientation but another radius, width or polarity leaves the decoded orientation intact in

Table 1: Pre-registered tests. IIA and normalized effect of the curve family, with 95% pigeonhole CIs, against the activation-change-matched null's q_{95}. Verdicts: H1 rows use the pre-registered three-way rule; the other rows use the secondary criterion (effect margin CI > 0, Holm $p < 0.05$). Full table: `reports/generated/results.md`.

Test	IIA family	null q_{95}	Effect (null q_{95})	Verdict
H1: 3b→4a curve units	0.177 [0.156, 0.199]	0.001	0.410 (0.098)	falsified
H1-ridge: 3b→all 4a	0.105 [0.087, 0.123]	0.000	0.248 (0.106)	supported
H2: curvature swap	0.405 [0.371, 0.440]	0.054	0.393 (0.141)	supported
H4: 3a→3b curve units	0.058 [0.045, 0.071]	0.005	0.324 (0.169)	supported
Composited (photos)	0.190 [0.169, 0.211]	0.001	0.415 (0.104)	supported
H5: natural rotation	0.072 [0.054, 0.091]	0.000	0.287 (0.075)	supported
H1, jitter seed 3	0.172 [0.151, 0.193]	0.001	0.410 (0.094)	falsified
H1, jitter seed 4	0.171 [0.149, 0.193]	0.001	0.407 (0.104)	falsified
H1, Lucent LRN	0.172 [0.150, 0.194]	0.000	0.406 (0.096)	falsified

at least 0.992 of stimuli, so the family does not inject nuisance information that the readout mistakes for orientation.

- **Stricter ablation null.** Rank-matched ablations retain the base orientation in 0.982 of pairs (family 0.806).

A third independent audit committed its overturning rules to git before running three checks. First, under the whole-layer ridge readout, ranking all 480 channels by their single-channel effect on the decoded $(\cos\theta, \sin\theta)$ puts 8 family units in the top 10, all within the top 24. The best 10 non-family channels reach 0.211 against the family's 0.573 (margin CI [0.350, 0.374]), so the specificity result does not rest on the partly circular curve-unit readout. It also withdrew the blur claim and narrowed the additivity claim, both described above.

5 Discussion

What the explanation gets right. The Circuits description of the `mixed3b` curve units survives every observational check we ran. Interventions add something observation cannot show: the units are causally *specific*. None of the 4000 sampled activation-change- and branch-matched 10-channel sets, nor (post hoc) the 1000 rank-matched sets or the 10 most-changing non-family channels, moves the next layer's curve code as far as the curve family does. A causal ranking of all 480 channels recovers 9 of the 10 published units as its top 10 for the curve-unit readout, and 8 for the whole-layer readout. The family also carries curvature evidence (H2), and removing it alters the network's decisions on natural images more than removing matched channels does.

What it gets wrong, or leaves out. The explanation is often read as "these neurons are where the network represents curve orientation". Under Huang et al.'s criterion, that reading fails: the family is not a sufficient mediator. The rest of the orientation signal is carried by many other `mixed3b` channels, about 80 for reliable transfer, and their contributions add almost exactly to the family's: exactly by architecture for 6 of the 14 readout units, and empirically for the rest. Whether those channels can replace the family depends on the readout: they do so for the next layer's curve units in 0.403 of pairs, but for a whole-layer decoder in only 0.019. This agrees with Gorton [15], who found curve features outside the 10 neurons, and with Huang et al. [19], whose

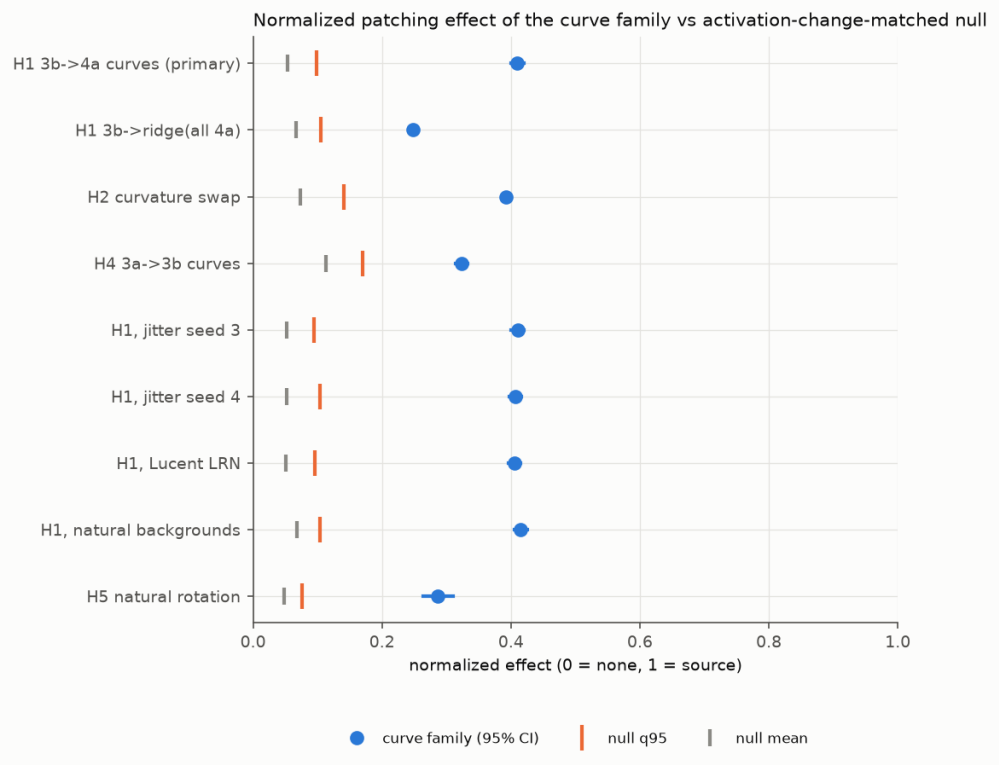


Figure 3: Normalized effect of the curve family (blue, 95% CI) for every test, against the Δ-matched null's mean (gray) and q_{95} (orange).

interchange accuracy in GPT-2 XL rose as larger fractions of neurons were intervened on. In a causal-abstraction framing [13], these results are consistent with an orientation variable realised by a distributed subspace of which the family is a privileged but partial part; we did not search for such a subspace.

Implications. Neuron-level explanations in vision should be stated with an explicit causal scope. Two claims need separating: "unit u is selective for X" (observational), and "unit set S mediates X for downstream computation Y" (interventional, with a stated readout). Activation-change-matched nulls are essential. Against random channels, almost any orientation-tuned set would look causally important, because most channels barely change between base and source.

6 Limitations

- **Single model and layer pair.** Only InceptionV1 and the 3b→4a (and 3a→3b) transition are tested. We do not claim generality to other CNNs or depths.
- **Partly circular readout.** The primary readout uses 4a curve units, which read `mixed3b` directly and plausibly use its curves: 3b curves feed 4a shape detectors, and 4a builds its own curve detectors mostly in its 5×5 branch [8]. We mitigate this with a whole-layer ridge readout and natural-image tests, but both decoders are trained on clean data and judge patched, possibly off-manifold states. A same-orientation sham keeps the readout intact in 0.999 of pairs, which argues against patching artefacts, but it does not rule out decoder bias.

- **Post-hoc analyses.** The rank-matched null, the causal competitor set, the per-direction H2 null, the cross-nuisance sham and the rank-matched ablation null were added after the confirmatory run and are exploratory. For four of the checks added by the second audit, the outcome that would overturn each claim was set before they ran, but git does not timestamp it. The third audit's three checks had their rules committed to git before they ran.
- **Strict binary metric.** IIA needs the decoded orientation within 10° of the source. Partial transfer appears only in the continuous effect. The pre-registered rule judges H1 on IIA, which makes it a statement about strong sufficiency.
- **Stimulus distribution.** The synthetic arcs are off-distribution for an ImageNet model. The composited and rotation experiments only partly address this. Rotating a photo changes far more than curve orientation, so H5's IIA is a conservative test.
- **No learned subspaces.** We test the published neuron set, not an optimised mediator such as distributed alignment search [12] or SAE features [15]. Whether a low-dimensional sufficient mediator exists is open.
- **Bootstrap assumptions.** The pigeonhole bootstrap treats base and source as independent factors, although every image plays both roles, and it ignores clustering within the 16 jitter cells. In H2, lines at θ and $\theta + 180°$ are near-duplicates. Our CIs may therefore be somewhat too narrow, although under a crossed random-effects model the pigeonhole bootstrap is itself mildly conservative [29]. This cannot change a verdict. H1 is falsified by its point estimate (IIA < 0.2): 0.177, with 0.171–0.172 for the other jitter seeds, although its CI upper bound (0.199) is close to 0.2. Every secondary margin's CI lower bound is at least 0.136.
- **Design details.** H5's clean decoding accuracy is 1.0 by construction, because its templates are the photo's own clean versions; success still requires the patched base to resemble the rotated version more than the unpatched base. The composited test reuses the same 32 backgrounds in both splits, so it generalises across jitter, not backgrounds. The 5% denominator filter for the effect was computed over all pairs rather than only eligible ones, as the pre-registration states. At most 1.7% of pairs are ineligible in any test (jitter seed 4), so the difference is negligible.
- **Pilot disclosure.** The pilot already suggested IIA ≈ 0.16. Thresholds were not changed, and the pilot is disclosed in the pre-registration. The effect-based secondary criterion, however, was specified after this pilot; the pilot did not compute the effect. Two smoke-test runs (20 null sets, tiny grids) before registration used about 99 of the 1962 confirmatory Imagenette photos, because the quick split did not respect the real halves (H5 IIA 0.029 on them); no rule was changed afterwards, and the split is now fixed.

7 Conclusion

Counterfactual concept swaps turn the Circuits curve-detector explanation into a falsifiable causal claim. The claim splits into two parts. The ten `mixed3b` curve units are a *specific* causal mediator of curve orientation and curvature: they far exceed activation-matched controls on synthetic and natural images and matter for the network's predictions. They are *not* a sufficient one: swapping them alone transfers the decoded orientation in only 0.177 of pairs, because the rest of the orientation signal is spread over many other channels. We release the pre-registration, code, per-pair data and figures so these numbers can be checked and extended.

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A Reproducibility and statistics details

Code, tests and the pre-registration are at https://github.com/BurnyCoder/counterfactually-causal-analysis-of-convolutional-neural-networks-with-concept-swaps. Per-pair data are in the Hugging Face dataset https://huggingface.co/datasets/BurnyCoder/inceptionv1-curve-concept-swaps. All numbers in this paper are LaTeX macros generated from the result files (reports/generated/numbers.tex).

Runtime is 57 minutes on an RTX 5070 Laptop GPU (8 GB), plus 22 minutes for the post-hoc checks, 5 minutes for the second audit's checks, 4 minutes for the third audit's and 3 minutes for the output-only example figures (Appendix C). Determinism: TF32 is disabled, and stimuli, pairs and null sets are seeded from one random stream in a fixed phase order, so only the full pipeline run reproduces the null sets exactly. Quick-mode runs of the commits that produced each result file and of the final code agree exactly in every shared field; the full confirmatory run was not repeated. The cropped-window recomputation equals the same block run on the full map in fp64 (unit test tolerance 10^{-9}; measured difference 0).

Null sizes (per Δ-matched and per branch-matched null): 2000 for H1 and H4; 1000 for H2, the composited test and H5; 500 for the ridge readout, the seeds and the LRN check. The descriptive random null has a quarter as many sets. The smallest attainable permutation p is $1/(M+1)$.

Deviations from the approved plan are listed in the pre-registration: the planned locality hypothesis became a unit test, the secondary decision criterion was specified, the stimulus radii were chosen on the pilot split, and the H5 decoder uses each photo's own clean versions as templates; all were decided before the confirmatory run. Deviations found afterwards: the effect's 5% denominator filter was computed over all pairs rather than eligible ones (Limitations; for H1 the effect changes from 0.40995 to 0.40996); the descriptive random null has a quarter of the matched nulls' size rather than 500; and quick-mode smoke runs used confirmatory-half Imagenette photos (Limitations). Three pre-registration errata are recorded in reports/all_experiments.md: the smoke-test use of confirmatory Imagenette photos above; the LRN constants follow TensorFlow's formula and a comment in the PR #51 discussion rather than the PR itself, and the 14 mixed4a units come from the authors' data file, whose article figure labels 12.

B Exploratory results

IIA by $|\Delta\theta|$ bin (30–60, 60–90, 90–120, 120–150, 150–180°): 0.111, 0.105, 0.222, 0.303, 0.149. H5 IIA by rotation angle: 0.030 at +45° and 0.140 at 180°.

C Example swaps

To show what the statistics count, the output-only examples phase draws six eligible H1 successes and three eligible failures uniformly at random (seed 0, not by outcome size). It shows each pair as it passes through the network (Figures 5–8). The phase recomputes the H1 per-pair outcomes and asserts that their IIA equals the confirmatory value. It also checks that swapping the family's 11×11 window in the full mixed3b maps and re-running mixed4a reproduces the scored readout up to float32 rounding. Following Cammarata et al. [7], each unit is shown by

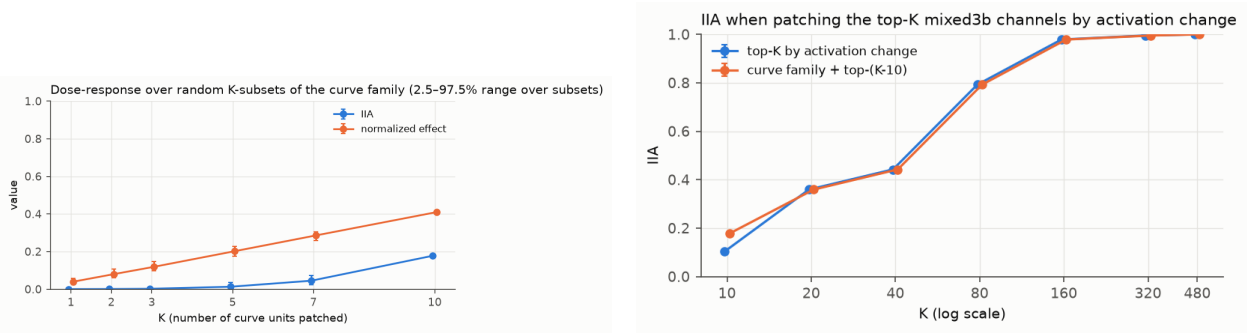


Figure 4: Left: dose-response over random K-subsets of the curve family. Right: IIA when patching the top-K `mixed3b` channels by activation change; the x axis shows $\log_2 K$.

a feature visualization [25] of the single neuron at the readout position, rendered with Lucent (`https://github.com/greentfrapp/lucent`). These images only illustrate the units; no analysis uses them, and feature visualizations can mislead [14]. In the failures shown, the swapped readout moves part of the way toward the source, but the decoder's peak stays near the base orientation. For the radius-80 arcs, curve-unit activity outside the swapped window stays the base's; it cannot affect the readout, which depends only on the window. These are single examples, not tests; the per-pair values are in `reports/generated/examples.md`.



Figure 5: Feature visualizations of the 10 `mixed3b` curve units (swapped) and the 14 `mixed4a` curve units (read), labelled with their E1 preferred orientation and cropped to the square holding 99% of the readout unit's input-gradient mass.

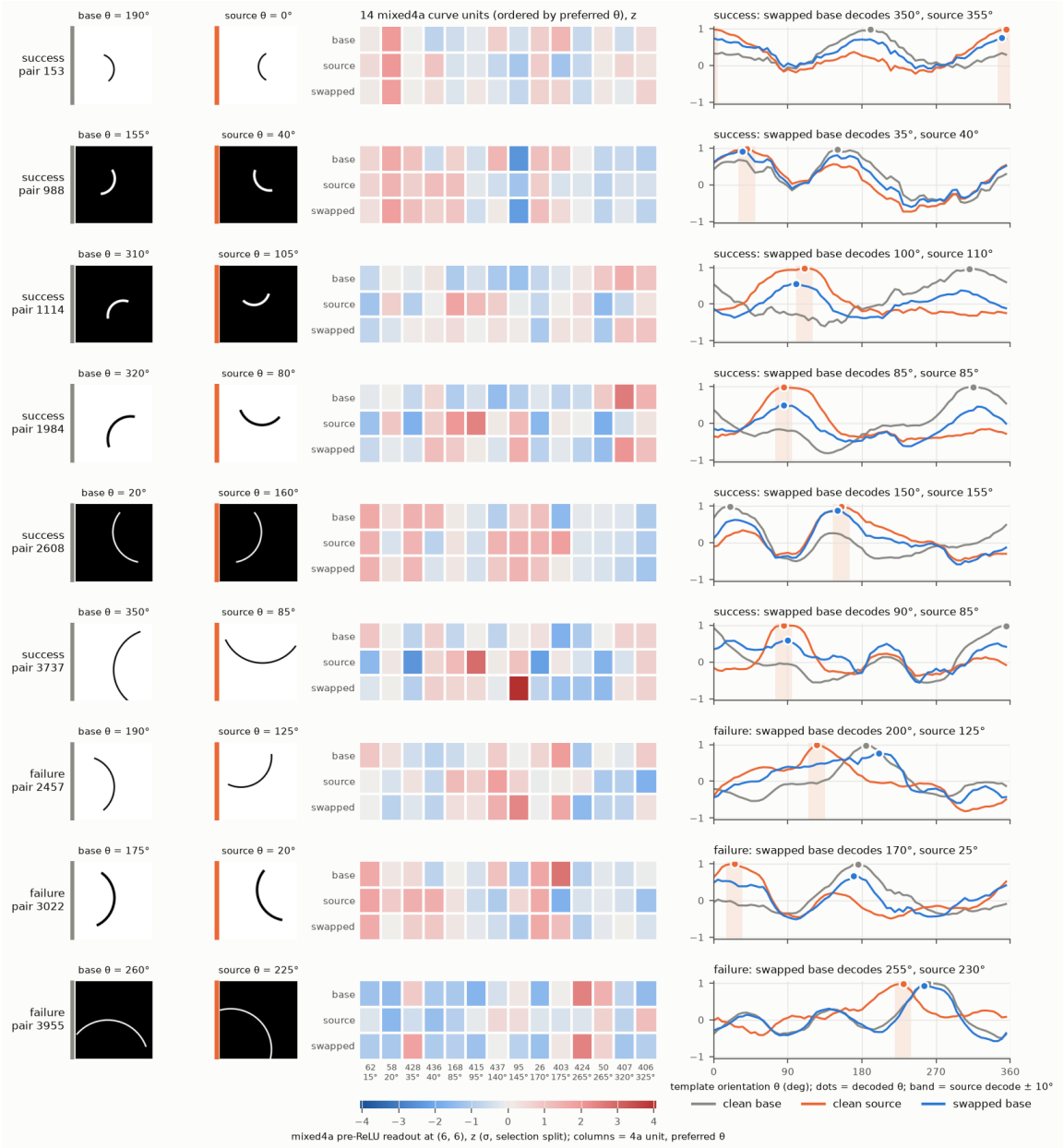


Figure 6: Six random H1 successes and three random failures. Per row: base and source inputs (central 152 px), the 14 `mixed4a` readouts (z) for clean base, clean source and swapped base, and the decoder's Pearson correlation with the 72 templates of the pair's cell. Dots mark decoded orientations; the band is the $\pm 10°$ success window around the source's decode.

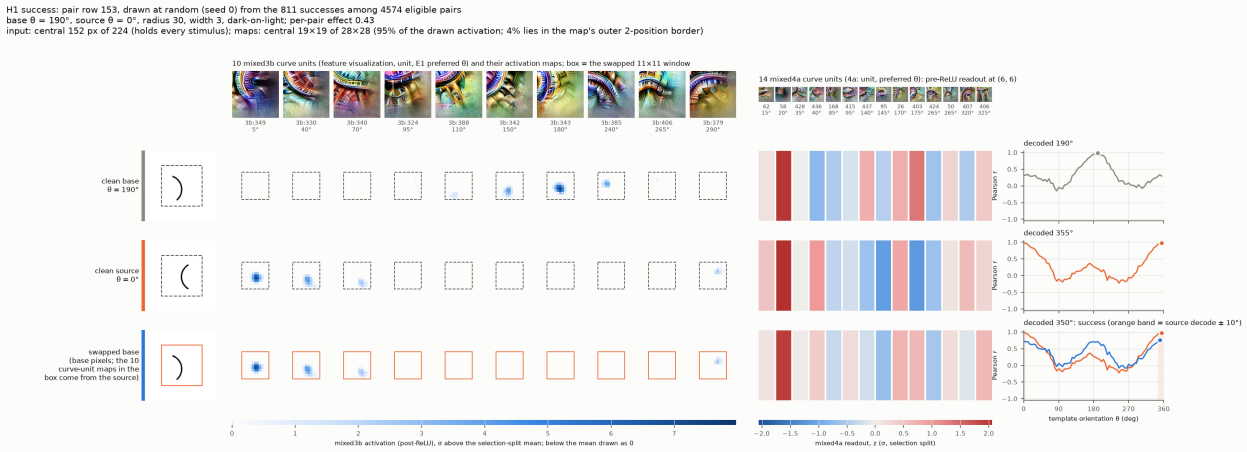


Figure 7: A random success (pair 153) flowing through the network. Rows: clean base, clean source, swapped base. Columns: input image, the 10 `mixed3b` curve-unit maps (σ above the selection-split mean) with the swapped 11×11 window boxed, the 14 `mixed4a` readouts, and the decoder output. In the swapped row the boxed maps are the source's.



Figure 8: A random failure (pair 2457), same layout. The readout moves toward the source, but the decoded orientation stays near the base's.

AI statement

This codebase and paper were written by Claude Code (Anthropic's coding agent, model Claude Opus 5.5). Libor Burian provided the exact research idea that Claude Code executed: counterfactual concept-swap interventions validating a Circuits neuron explanation in a CNN. Human cleanup and rewriting are in progress.